
ASSIGNMENTS

Problem set 2

AI504 — Knowledge Representation

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Problem 1

Work over the set $\{a, b, c, d\}$ of nouns. Consider the set of sentences

$$\Gamma = \{\text{All } a \text{ are } b, \text{All } a \text{ are } c, \text{All } b \text{ are } d, \text{All } c \text{ are } d\}.$$

Write down two different proof trees verifying $\Gamma \vdash \text{All } a \text{ are } d$. Write your proof trees carefully and as legibly as possible.

Solution

Here is the 1st solution

$$\frac{\text{all } a \text{ are } b \quad \text{all } b \text{ are } d}{\text{all } a \text{ are } d} \text{ BARBARA}$$

Figure 1: Proof tree 1

The Tree T on Figure 1 is composed of two different trees $\mathcal{T}_0$ (ie the left side of the T) and $\mathcal{T}_1$ (ie the right side of T). Using the BARBARA rule we show that T is a proof tree of Γ , this implies that $\Gamma \vdash \text{All } a \text{ are } d$.

Here is the 2nd solution

$$\frac{\text{all } a \text{ are } c \quad \text{all } c \text{ are } d}{\text{all } a \text{ are } d} \text{ BARBARA}$$

Figure 2: Proof tree 2

The Tree T on Figure 2 is composed of two different trees $\mathcal{T}_0^*$ (the left side of the T) and $\mathcal{T}_1^*$ (the right side of T). Using the BARBARA rule we show that T is a proof tree of Γ , this implies that $\Gamma \vdash \text{All } a \text{ are } d$.

Problem 2

Work over the set $\{a, b, c, d\}$ of nouns. In this question I want you to find **two different models of the same size**. Each of your models should satisfy all of the following sentences.

All a are b , All a are c , All b are d , All c are d and falsify both of the following sentences.
All b are c , All c are b .

Note. We have to be a little careful in saying what it means for two models to be different. But basically if you label the elements of your model $1, 2, \dots, n$ you should not be able to relabel (i.e., permute) the elements of one of your models to get the other one.

Solution

Let $\mathcal{M}$ be a model over the set of nouns $M = \{a, b, c, d\}$.

Model 1 ($\mathcal{M}_1$)

$$\begin{aligned}\llbracket a \rrbracket &= \emptyset \\ \llbracket b \rrbracket &= \{b\} \\ \llbracket c \rrbracket &= \{c\} \\ \llbracket d \rrbracket &= \{b, c, d\}\end{aligned}$$

This model would then **satisfy** the following:

$$\mathcal{M}_1 \models \text{all } a \text{ are } b$$

$$\mathcal{M}_1 \models \text{all } a \text{ are } c$$

$$\mathcal{M}_1 \models \text{all } b \text{ are } c$$

$$\mathcal{M}_1 \models \text{all } c \text{ are } d$$

And **falsify**:

$$\mathcal{M}_1 \not\models \text{all } b \text{ are } c$$

$$\mathcal{M}_1 \not\models \text{all } c \text{ are } b$$

Model 2

Let $\mathcal{M}$ be a model where $M = \{a, b, c, d\}$.

$$\llbracket a \rrbracket = \{a\}$$

$$\llbracket b \rrbracket = \{a, b\}$$

$$\llbracket c \rrbracket = \{a, c\}$$

$$\llbracket d \rrbracket = \{a, b, c, d\}$$

This model would then **satisfy** the following:

$$\mathcal{M}_2 \models \text{All } a \text{ are } b$$

$$\mathcal{M}_2 \models \text{All } a \text{ are } c$$

$$\mathcal{M}_2 \models \text{All } b \text{ are } d$$

$$\mathcal{M}_2 \models \text{All } c \text{ are } d$$

And **falsify**:

$$\mathcal{M}_2 \not\models \text{All } b \text{ are } c$$

$$\mathcal{M}_2 \not\models \text{All } c \text{ are } b$$

Problem 3

The *dual* of a sentence is obtained by switching it's two nouns. For example the dual of all p are q is all q are p . We denote the dual of a sentence ϕ by $\phi^\dagger$. If Γ is a set of sentences, by $\Gamma^\dagger$ we mean the obvious thing, i.e., "take the dual of each sentence in Γ ."

Suppose $\Gamma \vdash \phi$. Prove that $\Gamma^\dagger \vdash \phi^\dagger$ by induction on proof trees. Be very careful and deliberate, so that I know you understand what you're doing.

Solution

Proof by induction

Goal: if T is a proof of ϕ from Γ then, there exist a proof such that $\Gamma^\dagger \vdash \phi^\dagger$.

Base case (T is a single leaf)

1 Suppose T is a proof of ϕ from Γ . Since T is a single leaf, the tree is just ϕ , and since all leaves are either premises and come from Γ , or is on the form "all p are p ", which is an axiom, it follows that $\phi \in \Gamma$, or $\phi = \phi^\dagger$ and thus is trivial.

2 For the dual $\phi^\dagger$ from ϕ , if $\phi \in \Gamma$ and $\Gamma^\dagger$ contains all the duals of Γ , then $\phi^\dagger \in \Gamma^\dagger$.

Inductive hypothesis

3 For all proofs of T , Γ , and ϕ , if T is a proof of ϕ from Γ , then there exist a proof T , such that $\Gamma^\dagger \vdash \phi^\dagger$

Inductive step

4 Suppose T is not a leaf, decompose T as follows.

5 Let's assume that $\mathcal{T}$ is not a single leaf ϕ and $\phi^\dagger$, but $\mathcal{T}$ is composed of two trees, left and right trees labeled $\mathcal{T}_0$ and $\mathcal{T}_1$, respectively, and it's dual:

$$6 \quad \frac{\mathcal{T}_0 \quad \mathcal{T}_1}{\text{all } p \text{ are } x \quad \text{all } x \text{ are } q} \text{ BARBARA} \\ \text{all } p \text{ are } q$$

Figure 3: Proof tree 3

7 The proof tree can then be restructured, so it satisfies the $\phi^\dagger$ dual, which will then satisfy

$$8 \quad \frac{\mathcal{T}_1^\dagger \quad \mathcal{T}_0^\dagger}{\text{all } q \text{ are } x \quad \text{all } x \text{ are } p} \text{ BARBARA} \\ \text{all } q \text{ are } p$$

Figure 4: $T^\dagger$

9 By the basis of the inductive hypothesis applied to $\mathcal{T}_0$ and $\mathcal{T}_1$, it follows that $\Gamma^\dagger \vdash \mathcal{T}_0^\dagger$ and $\mathcal{T}_1^\dagger$, and by using the BARBARA rule, it follows that $\Gamma^\dagger \vdash \phi^\dagger$.

Problem 4

Continuing the previous problem, there is also a notion of a dual model. So the idea is that given a model $\mathcal{M}$, we should be able to define its “dual” $\mathcal{M}^\dagger$, which has the property that $\mathcal{M}^\dagger \models \phi^\dagger$ if and only if $\mathcal{M} \models \phi$, for each sentence ϕ . Define the dual and prove that it enjoys this property (Hint. Start out with some explicit small models, see if you can find their duals by trial and error, and then see what the common pattern is.)

Solution

First lets define $\mathcal{M}$ and $\mathcal{M}^\dagger$

$$\mathcal{M} = (M, \llbracket \cdot \rrbracket, M \rightarrow \mathcal{P}(M))$$

$$\mathcal{M}^\dagger = (M, \llbracket \cdot \rrbracket^\dagger, M \rightarrow \mathcal{P}(M))$$

By observing some examples we see that the dual interpret can be described as

$$\llbracket p \rrbracket^\dagger = f(\llbracket p \rrbracket) = M \setminus \llbracket p \rrbracket$$

in a case of $\llbracket p \rrbracket \subseteq \llbracket q \rrbracket$ that means that $f(\llbracket q \rrbracket) \subseteq f(\llbracket p \rrbracket)$ (which will be proved below).

We can see this in this example on Figure 5 where $\llbracket p \rrbracket = \{1\} \subseteq \llbracket q \rrbracket = \{1, 2\} \subseteq M$

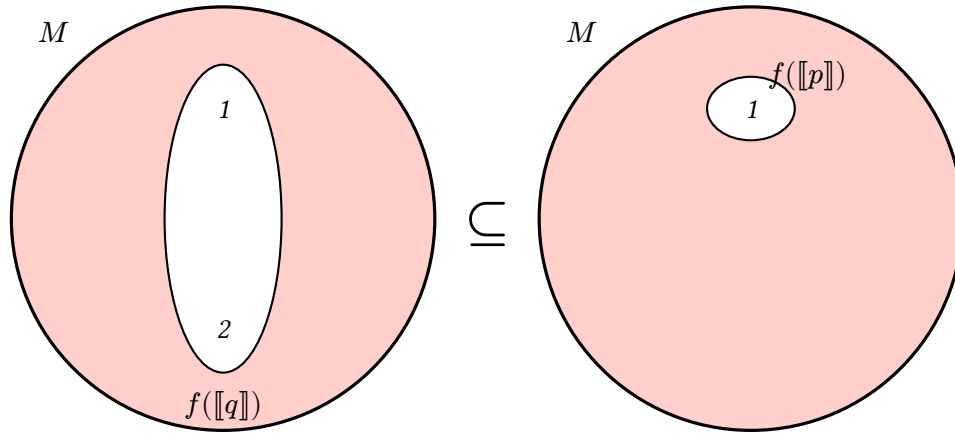


Figure 5: $\llbracket p \rrbracket \subseteq \llbracket q \rrbracket \subseteq M \Rightarrow f(\llbracket q \rrbracket) \subseteq f(\llbracket p \rrbracket) = M \setminus \llbracket q \rrbracket \subseteq M \setminus \llbracket p \rrbracket$

Now we can use the interpretations $\llbracket p \rrbracket$ and $\llbracket q \rrbracket$

If $\llbracket p \rrbracket \subseteq \llbracket q \rrbracket$ that means that $f(\llbracket q \rrbracket) \subseteq f(\llbracket p \rrbracket)$

If $\llbracket p \rrbracket \subseteq \llbracket q \rrbracket$ that means that $\llbracket q \rrbracket^\dagger \subseteq \llbracket p \rrbracket^\dagger$

This is the same as

$$\llbracket q \rrbracket^\dagger \subseteq \llbracket p \rrbracket^\dagger \text{ iff } \llbracket p \rrbracket \subseteq \llbracket q \rrbracket$$

Which is

$$\mathcal{M}^\dagger \models \phi^\dagger \text{ iff } \mathcal{M} \models \phi$$

PROOF

We want to prove that:

$$x \subseteq y \iff M \setminus y \subseteq M \setminus x$$

We begin first with proving the one way implication:

$$x \subseteq y \implies M \setminus y \subseteq M \setminus x$$

Where:

$$x \subseteq y \equiv \forall z \in x : z \in y$$

And:

$$M \setminus y \equiv \forall z : z \in M \setminus y \iff z \in M \text{ and } z \notin y$$

$$M \setminus x \equiv \forall z : z \in M \setminus x \iff z \in M \text{ and } z \notin x$$

We start by assuming $x \subseteq y$. Let $z \in M \setminus y$. We have to prove:

$$z \in M \setminus x$$

We then need to figure out if it is true that:

$$z \notin x$$

We can assume that $z \in x$ and attempt to get to a contradiction. We know that if $z \in x$ then $z \in y$ from earlier, since x is a subset of y , and therefore $z \in y$ would need to be true for the $z \in x$ to hold, but we know $z \in M \setminus y \iff z \in M$ and $z \notin y$. Where it is stated that z is not in y , that means z cannot be in x either (because $x \subseteq y$), meaning that:

$$z \notin x$$

This completes the one way implication. Then we need to prove the other direction for the entire bi-implication to hold. We attempt to prove:

$$M \setminus y \subseteq M \setminus x \implies x \subseteq y$$

We start by assuming $M \setminus y \subseteq M \setminus x$. To prove that x is a subset of y , assume $z \in x$ but $z \notin y$. If $z \in x$, $z \in M$. This ensures that $z \in M \setminus y$. That further shows that $z \in M \setminus x$. If $z \in M \setminus x$, then $z \notin x$. This shows a contradiction, therefore we can conclude that the implication holds.

This means that the bi-implication holds.

Problem 5

The point of this exercise is to show you that models can be built out of anything, as long as they have the correct type. Work over the set $N = \{0, 1, 2, 3\}$ of nouns. Now this set of nouns makes for weird-looking sentences like “all 3 are 1”. But once we construct an N -model, it makes perfect sense to say that such sentences are satisfied or falsified.

Define the N -model $\mathcal{R}$ as follows. Its domain is $\mathbb{R}$, the set of real numbers, and for each $b \in N$, define

$$\llbracket b \rrbracket = \{x \in \mathbb{R} : x^b - x < 0\}$$

List all of the N -sentences that $\mathcal{R}$ satisfies, and argue that your answer is correct. (For the purposes of this exercise, assume $0^0 = 1$.)

Solution

To show this, we can construct all the nouns $\llbracket b \rrbracket$ from N that follows $\mathcal{R}$:

$$b = 0 \implies \{x \in \mathbb{R} : x^0 - x < 0\} \implies \{x \in \mathbb{R} : 1 - x < 0\} \implies \{x \in \mathbb{R} : x > 1\}$$

$$b = 1 \implies \{x \in \mathbb{R} : x^1 - x < 0\} \implies \{x \in \mathbb{R} : x - x < 0\} \implies \{x \in \mathbb{R} : x < x\}$$

$$b = 2 \implies \{x \in \mathbb{R} : x^2 - x < 0\}$$

$$b = 3 \implies \{x \in \mathbb{R} : x^3 - x < 0\}$$

Then, we can for each of the constructed nouns attempt to find the places where the values are true according to the model $\mathcal{R}$.

For $b = 0$, this one becomes obvious due to its simple nature. We know that all values have to be larger than 1, so we simply get:

$$b = 0 \implies \{x \in \mathbb{R} : x > 1\} \implies (1, \infty)$$

$$\llbracket 0 \rrbracket = (1, \infty)$$

$b = 1$, there is an inequality that can never be satisfied, so no x will be allowed in the set, making it the empty set $\emptyset$:

$$b = 1 \implies \{x \in \mathbb{R} : x < x\} \implies (\emptyset)$$

$$\llbracket 1 \rrbracket = \emptyset$$

Then for $b = 2$, the value must be calculated, as the solution is not as intuitive as the previous 2, so we try to find every value of x , where $y = 0$, since this is the cutoff according to the inequality:

$$b = 2 \implies \{x \in \mathbb{R} : x^2 - x < 0\}$$

Making the equation:

$$x^2 - x = 0$$

We use the quadratic formula $x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$ to find the solutions, and calculate the discriminant $d = b^2 - 4ac$ shows, which shows there will be 2 solutions, since it is positive.

$$d = (-1)^2 - 4 \cdot 1 \cdot 0 = 1$$

The quadratic formula will then take both a positive and negative form to find the x 's:

$$x_1 = \frac{-(-1) + \sqrt{(-1)^2 - 4 \cdot 1 \cdot 0}}{2 \cdot 1} = \frac{1 + \sqrt{1}}{2} = \frac{2}{2} = 1$$

$$x_2 = \frac{-(-1) - \sqrt{(-1)^2 - 4 \cdot 1 \cdot 0}}{2 \cdot 1} = \frac{1 - \sqrt{1}}{2} = \frac{0}{2} = 0$$

Now we just need to know where $y < 0$ so we can find the interval. A quick look at the formula would show that it is convex, so we calculate $x = 0.5$ to make sure that it is between the two points, that are negative.

$$x^2 - x \rightarrow 0.5^2 - 0.5 = -0.25$$

So the interval becomes:

$$\llbracket 2 \rrbracket = (0, 1)$$

$b = 3$, once again the equation can be solved, for whenever it is below 0, since this value is given by the inequality. Therefore, we attempt to find all x where $y = 0$.

$$b = 2 \implies \{x \in \mathbb{R} : x^3 - x < 0\}$$

We attempt using the zero product property to find the correct x 's. We start by factoring out the most common factor x , so we get:

$$x^3 - x = x \cdot (x^2 - 1)$$

Then we can apply the difference of squares rule, since this can be written as:

$$x \cdot (x^2 - 1) = x \cdot (x^2 - 1^2) = x \cdot (x - 1) \cdot (x + 1)$$

Then we can apply the zero product property and solve each product by themselves, as they each have their own x :

$$x = 0$$

$$x - 1 = 0 \implies x = 1$$

$$x + 1 = 0 \implies x = -1$$

So we now know all the places where $y = 0$, the missing step is to find intervals where $y < 0$:

$$x = -2 : x^3 - x \rightarrow -2^3 - (-2) = -6$$

$$x = -0.5 : x^3 - x \rightarrow -0.5^3 - (-0.5) = 0.375$$

$$x = 0.5 : x^3 - x \rightarrow 0.5^3 - 0.5 = -0.375$$

$$x = 2 : x^3 - x \rightarrow 2^3 - 2 = 6$$

We can then observe that before and until $x = -1$ the values are in the negative, changing to positive and then back to negative at $x = 0$, and then back to positive at $x = 1$. Making the interval where the equation is fulfilled according to the inequality:

$$\llbracket 3 \rrbracket = (-\infty, -1) \cup (0, 1)$$

Thereby, we know that all the correct intervals for the values fall at:

$$\llbracket 0 \rrbracket = (1, \infty)$$

$$\llbracket 1 \rrbracket = (\emptyset)$$

$$\llbracket 2 \rrbracket = (0, 1)$$

$$\llbracket 3 \rrbracket = (-\infty, -1) \cup (0, 1)$$

Therefore, the list of the N-sentences that $\mathcal{R}$ satisfies:

$$\mathcal{R} \models \text{All of 2 are 3} \Rightarrow \llbracket 2 \rrbracket \subseteq \llbracket 3 \rrbracket$$

$$\mathcal{R} \models \text{All of 1 are 0} \Rightarrow \llbracket 1 \rrbracket \subseteq \llbracket 0 \rrbracket$$

$$\mathcal{R} \models \text{All of 1 are 2} \Rightarrow \llbracket 1 \rrbracket \subseteq \llbracket 2 \rrbracket$$

$$\mathcal{R} \models \text{All of 1 are 3} \Rightarrow \llbracket 1 \rrbracket \subseteq \llbracket 3 \rrbracket$$